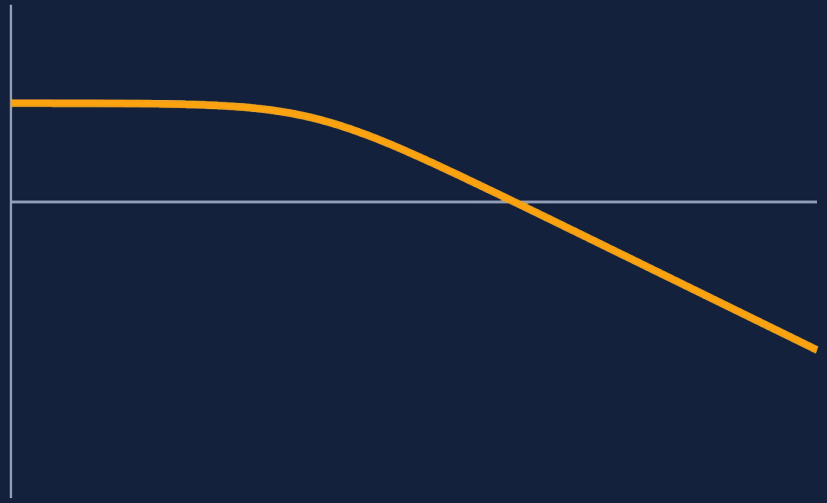
Bode Plot, Gain and Phase Margin



Control Systems (22EEG53)

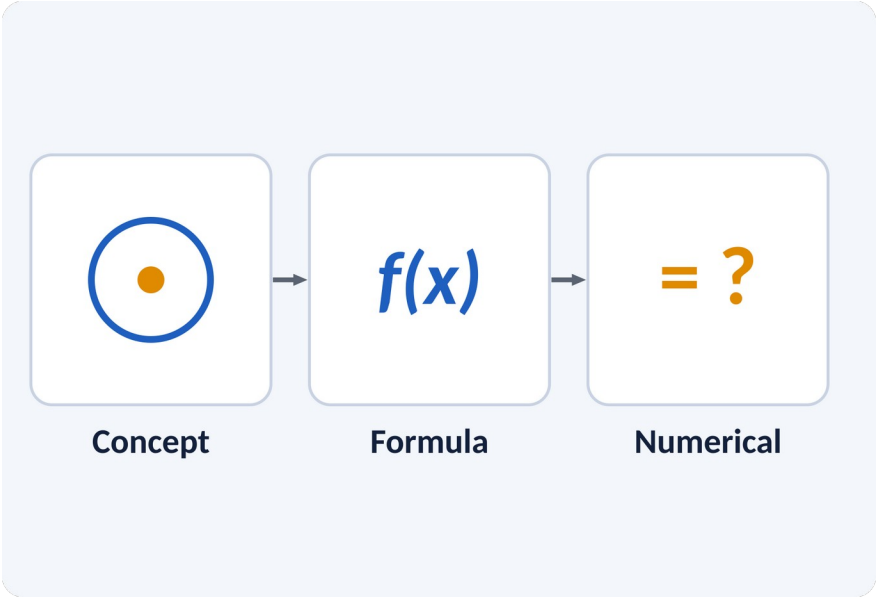
Unit 3: Frequency Domain Analysis

Topic: Bode plot construction; gain margin and phase margin

Prof. Mitesh Astik

Learning Outcomes

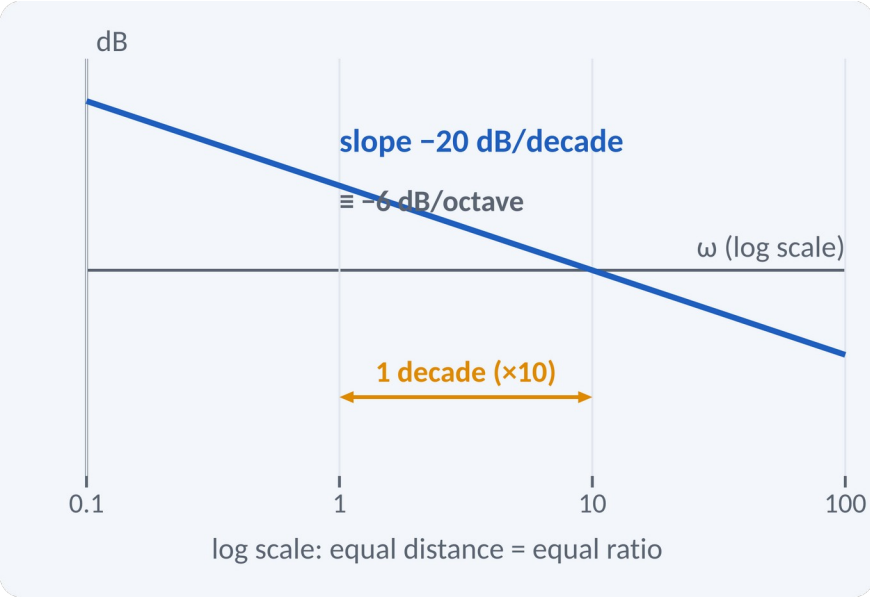
- Draw the asymptotic Bode plot of G(jω).
- Locate gain and phase crossover frequencies.
- Find gain margin and phase margin.
- Judge relative stability and estimate ζ.



Linked to CO3
Analyse performance indices of a system in the frequency domain.

Recap: What You Already Know

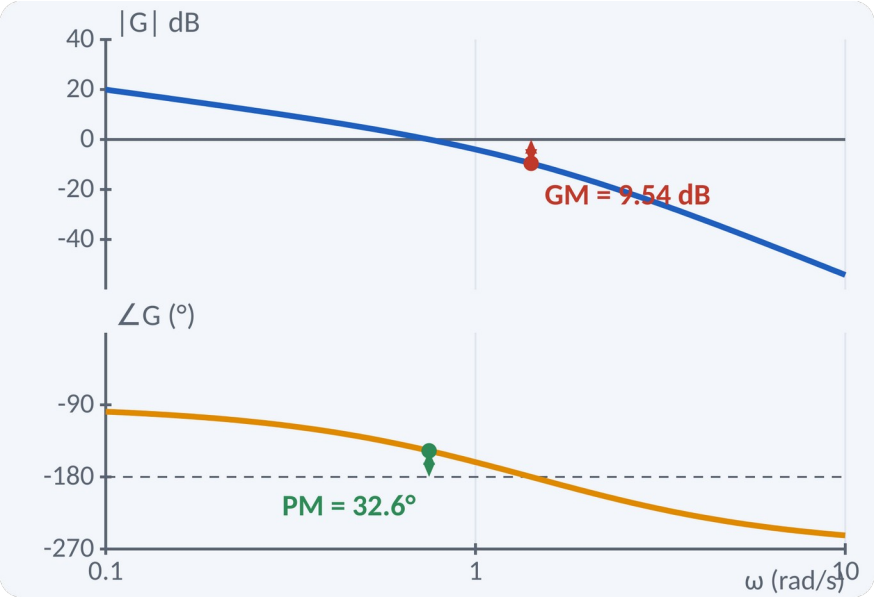
- Frequency response G(jω): magnitude and phase.
- Decibel: dB = 20 $\log_{10}|G|$.
- Log scale: equal distance = equal ratio.
- Factor G(s) into simple terms.



$|G1 \cdot G2|$ in dB = $|G1|$ dB + $|G2|$ dB; $\angle(G1 \cdot G2) = \angle G1 + \angle G2$
decade = ×10 in ω; octave = ×2

Motivation: How Close to Instability?

- Too much gain makes a loop oscillate.
- Margins say how much gain or phase is spare.
- Margins are read directly from a Bode plot.
- No need to compute closed-loop poles.



Gain margin and phase margin = safety factors of a feedback design

Bode Plot of Standard Factors

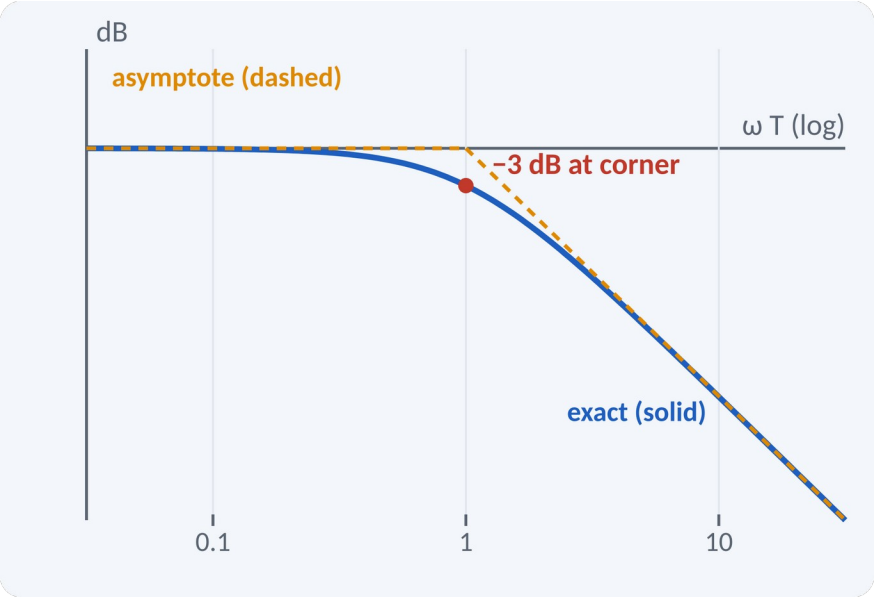
- Constant K: flat at 20 log K dB.
- Integrator 1/s: −20 dB/dec, −90°.
- First-order lag: slope changes by −20 at 1/T.
- Phase moves −45° at the corner frequency.
- Zero factor (1 + sT) mirrors the lag.

Factor	Magnitude slope	Phase
K	flat, 20 log K	0°
1 / s	−20 dB/dec	−90°
s	+20 dB/dec	+90°
1/(1 + sT)	0 → −20 after 1/T	0° → −90°
1 + sT	0 → +20 after 1/T	0° → +90°

Corner frequency $\omega c = 1/T$; $|1/(1 + j\omega T)| = -3$ dB at ωc

Asymptotic and Exact Plot

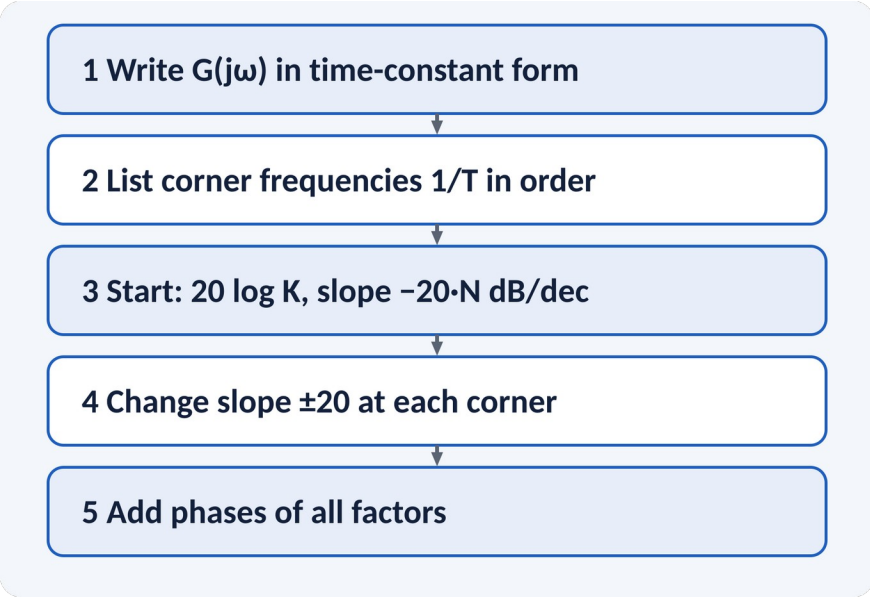
- Asymptotes: two straight lines meeting at ωc.
- Exact curve lies below by 3 dB at the corner.
- Error is 1 dB at ωc/2 and 2ωc.
- Asymptotic plot is enough for design sketches.



Exact: $-10\log_{10}(1+\omega^2T^2)$; asymptote: 0 dB then −20 dB/dec
Maximum error 3 dB at ω = 1/T

Steps to Draw a Bode Plot

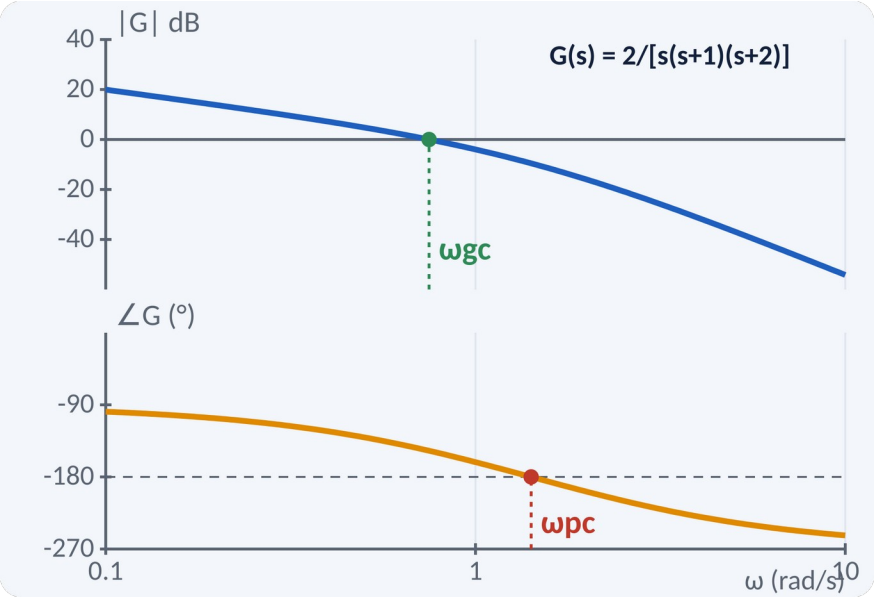
- Convert G(s) to the time-constant form.
- Mark corner frequencies on the log axis.
- Start with the low-frequency asymptote.
- Change the slope at every corner.
- Add the phase of all factors.



Low-frequency asymptote: slope = −20N dB/dec, passes 20 log K at ω = 1 (N integrators)

Gain and Phase Crossover Frequencies

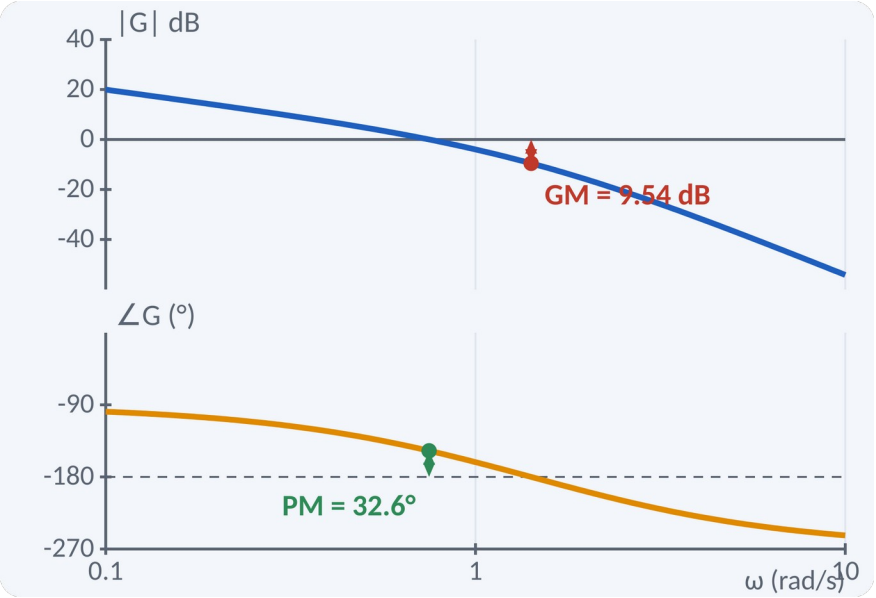
- Gain crossover ωgc: |G(jω)| = 1 (0 dB).
- Phase crossover ωpc: ∠G(jω) = −180°.
- Both are read from the open-loop Bode plot.
- Stable closed loop: ωgc < ωpc.



|G(jωgc)| = 1; ∠G(jωpc) = −180°
Example G(s) = 2/[s(s + 1)(s + 2)]: ωgc = 0.749, ωpc = 1.414 rad/s

Gain Margin and Phase Margin

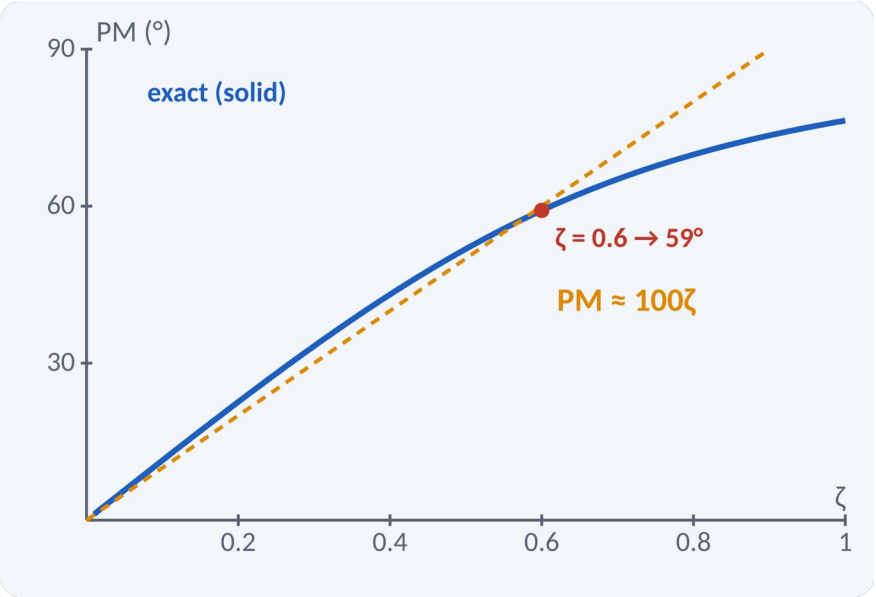
- GM: extra gain allowed at ωpc before instability.
- PM: extra phase lag allowed at ωgc.
- GM > 0 dB and PM > 0° mean a stable loop.
- Typical design: GM ≥ 6 dB, PM 30°–60°.



GM = 1/|G(jωpc)| (or −20 log|G(jωpc)| dB); PM = 180° + ∠G(jωgc)
GM dimensionless or dB, PM in degrees

Phase Margin and Damping Ratio

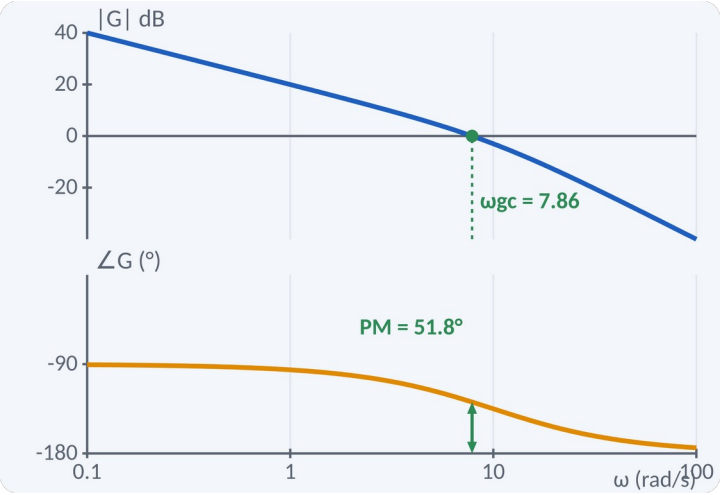
- Phase margin relates to damping of the closed loop.
- Larger PM means smaller overshoot.
- Rule of thumb: ζ ≈ PM/100 (PM below about 60°).
- PM = 60° corresponds to ζ ≈ 0.6.



$PM = \tan^{-1}\{ 2\zeta / \sqrt{(\sqrt{(1 + 4\zeta^4)} - 2\zeta^2)} \} \approx 100\zeta$
For a second-order system with $G = \omega_n^2/[s(s + 2\zeta\omega_n)]$

Solved Example 1: Bode of 10/[s(1+0.1s)]

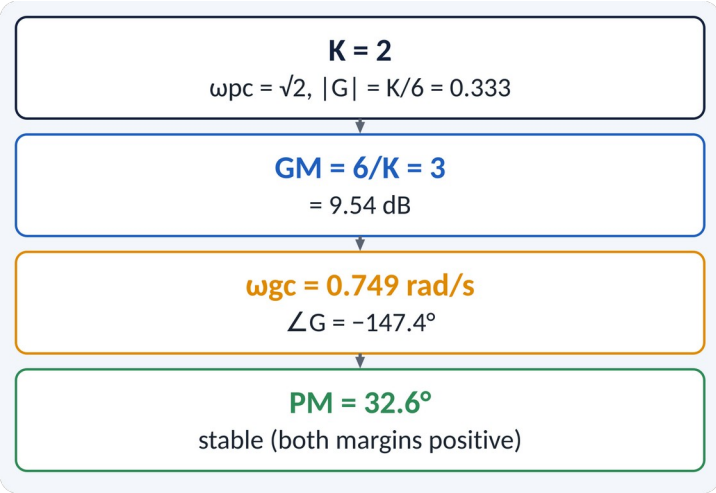
- Draw asymptotic Bode plot; find ωgc, GM and PM for G = 10/[s(1 + 0.1s)].
- 1. One integrator: slope −20 dB/dec; 20 log 10 = 20 dB at ω = 1.
- 2. Corner at 1/0.1 = 10 rad/s: slope becomes −40 dB/dec.
- 3. ωgc: $100 = \omega^2(1 + 0.01\omega^2) \rightarrow \omega gc = 7.86$ rad/s.
- 4. $\angle G = -90° - \tan^{-1}(0.1\omega) = -128.2°$ at ωgc → PM = 51.8°.
- 5. Phase never reaches −180°: GM = ∞.



Answer: ωgc = 7.86 rad/s, PM = 51.8°, GM = ∞ (always stable)

Solved Example 2: GM and PM

- Open loop G = K/[s(s + 1)(s + 2)] with K = 2. Find ωpc, GM, ωgc, PM, and the K giving marginal stability.
- 1. Phase = −90° − $\tan^{-1}\omega$ − $\tan^{-1}(\omega/2)$ = −180° → ωpc = √2 = 1.414.
- 2. |G(jωpc)| = K/(√2·√3·√6) = K/6 = 0.333.
- 3. GM = 6/K = 3 = 9.54 dB.
- 4. |G| = 1 at ωgc = 0.749; ∠G = −90 − 36.9 − 20.6 = −147.4°.
- 5. PM = 180 − 147.4 = 32.6°.
- 6. Marginal stability when GM = 1: K = 6.



Answer: GM = 9.54 dB, PM = 32.6°, stable; marginal at K = 6

Common Mistakes and Exam Tips

Common mistakes

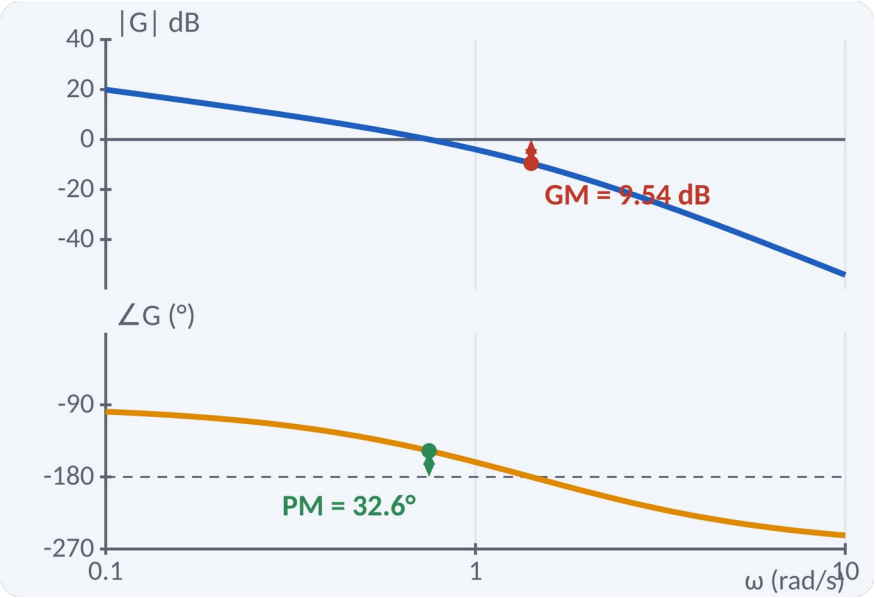
- Forgetting the sign: PM = 180° + ∠G, not 180° − ∠G.
- Reading margins from the closed-loop plot.
- Using natural log instead of $\log_{10}$ in dB.
- Putting the corner at T instead of 1/T.
- Concluding stability from GM alone.

Exam tips

- Write G(jω) in time-constant form.
- Check both ωgc < ωpc and PM > 0.
- Sketch asymptotes first, then correct.
- State GM in dB or ratio clearly.

Summary

- Bode: magnitude (dB) and phase against log ω.
- Each pole/zero changes slope by ∓20 dB/decade.
- ωgc: $|G| = 1$; ωpc: $\angle G = -180°$.
- GM = $1/|G(j\omega_{pc})|$; PM = $180° + \angle G(j\omega_{gc})$.
- PM ≈ 100ζ gives a quick estimate of damping.



Stable loop: GM > 0 dB and PM > 0° (minimum-phase)

Quick Quiz

Q1. A simple integrator 1/s has Bode slope:
(a) +20 (b) −20 (c) −40 (d) 0 dB/dec

Ans: (b)

Q2. The phase margin is measured at:
(a) ωpc (b) ωgc (c) ω = 0 (d) ωn

Ans: (b)

Q3. Gain margin is 1/|G| at the:
(a) gain crossover (b) phase crossover (c) corner (d) peak

Ans: (b)

Q4. |G(jωpc)| = 0.25 gives a GM of:
(a) 2 (b) 4 (c) 0.25 (d) 12 dB only

Ans: (b)

Q5. PM of 45° suggests ζ about:
(a) 0.1 (b) 0.45 (c) 0.9 (d) 1.45

Ans: (b)

References

1. **Benjamin C. Kuo, Automatic Control Systems**
   7th Ed., PHI, 2007 (Units 3, 4, 5 textbook)

2. **P. S. Satyanarayana, Concepts of Control Systems**
   2nd Ed., Dynaram Publishers, Bangalore, 2001 (reference)

3. **K. Ogata, Modern Control Engineering**
   5th Ed., Pearson Education Asia / PHI, 2009 (reference)

Next lecture: Unit 4: stability and the Routh–Hurwitz criterion